

Wright State University
Computer Science and Engineering

CS4710-6710– Assignment 3
Introduction to Data Mining

Instructor: Dr. Tomojit Ghosh

Due by 2025/11/30

This assignment is mandatory for section CS6710 and optional for CS4710 . Good luck !

Distribution of Marks

Question	Points	Score
1	70	
2	30	
Total:	100	

- (70 points) **Objective:** In this question you will implement and test the Exact and Naive Bayes classifier on Zoo data and WeatherAndPlayData. Split each data into train and test partition using 80 : 20 ratio. While splitting the Zoo data make sure you have atleast one sample from each class in your test partition as the distribution of samples is not uniform in this dataset. You need to run 10 repetition and report average accuracy with standard deviation. Also compare the result with linear regression classifier. So the comparison will be made on Exact Bayes vs Naive Bayes vs Linear Regression classifier.
- (30 points) **Objective:** In this question you will analyze the Gini index on the above mentioned

Customer Id	Gender	Car Type	Shirt Size	Class
1	M	Family	Small	C0
2	M	Sports	Medium	C0
3	M	Sports	Medium	C0
4	M	Sports	Large	C0
5	M	Sports	Extra Large	C0
6	M	Sports	Extra Large	C0
7	F	Sports	Small	C0
8	F	Sports	Small	C0
9	F	Sports	Medium	C0
10	F	Luxury	Large	C0
11	M	Family	Large	C1
12	M	Family	Extra Large	C1
13	M	Family	Medium	C1
14	M	Luxury	Extra Large	C1
15	F	Luxury	Small	C1
16	F	Luxury	Small	C1
17	F	Luxury	Medium	C1
18	F	Luxury	Medium	C1
19	F	Luxury	Medium	C1
20	F	Luxury	Large	C1

data set which was discussed in the class in the context of Information Gain, see the slides from (*CS4710-6710_Topic4.ClassificationAndModelEvaluation_part2.pdf*). Split the data on gender, car type and customer id. Which attribute gave you the maximum gain? Do you see any issue with this partitioning approach? If yes can you come up with an improved approach.

Submission Instructions:

- You submission must contain a pdf file, python scripts. The report (pdf file) must contain a discussion on the results.
- Submit all files via the course portal by **November 30, 2025**.
- Late submissions will incur penalties as per course policy.

Note:

- You are allowed to use Panda to load data set.
- Do not use any machine learning libraries such as `scikit-learn`.
- You may use `numpy` and `matplotlib` for numerical computation and visualization.
- You are welcome to discuss ideas with your peers, but your submitted code and analysis must be entirely your own work.